EXPERIMENT 4

Demonstration of Network Communication Models (OSI / TCP-IP) Using a Network Simulation Tool

Subject: No-Code Application Development

Aim

To demonstrate network communication between two hosts and relate observed packet exchanges to the OSI and TCP/IP models using Cisco Packet Tracer or an equivalent simulator.

Objectives

- Identify the layers and functions of OSI and TCP/IP models.
- Create a simple local-area network and configure host IP addresses.
- Generate test traffic and inspect encapsulation as packets move through the network.

Selected setup

A simulated LAN containing two PCs connected through a switch. PC0: 192.168.1.10/24; PC1: 192.168.1.20/24. Both hosts belong to the same subnet; no router is required for their communication.

Requirements / Tools

- Cisco Packet Tracer (or a comparable network simulator)
- Two end devices, one switch, and two copper straight-through links
- IP addressing plan and access to simulation/event view

No-Code Connection

A no-code web app still relies on TCP/IP networking. Its visual form builder hides implementation details but the published application, API requests, and database access still require reliable transport, addressing, and security.

Illustrative Business Scenario

A no-code inventory dashboard can be opened on a phone and laptop. Use the simulated LAN to explain why both devices need correct connectivity before app data can synchronize.

Experimental Procedure

Follow these steps in the designated simulator, browser or visual app builder. Menu locations and available features may vary by product version and account plan.

1. Place two PCs and one switch in the workspace. Connect each PC to a switch port.
2. Assign the stated IPv4 addresses and subnet mask 255.255.255.0 to the PCs.
3. From PC0, ping 192.168.1.20 and record success or failure.
4. Switch to Simulation mode; filter ARP and ICMP events and repeat the ping.
5. Open a packet event, examine the Ethernet frame and IP/ICMP data, and note each encapsulation step.
6. Optionally add a router and second network to observe the role of Layer 3 routing.

No-Code Application Perspective

- Identify the user action and the visual interface element involved (form, page, workflow or report).
- Trace the data movement from input to processing to stored record, report or notification.
- Identify configuration that replaces traditional hand-written code and note where scripting might still be required.
- Check whether the experiment can be repeated using sample records and documented settings.

Evidence to Collect

- Screenshot of the configured screen, topology, tool or application.
- A short observation table showing input, action and output.
- One successful test and one failed or exceptional test with expected behavior.

OSI and TCP/IP layer mapping

OSI layer	Main function	TCP/IP grouping
7 Application	User-facing protocols	Application
6 Presentation	Encoding and formatting	Application
5 Session	Session management	Application
4 Transport	End-to-end delivery	Transport
3 Network	Logical addressing and routing	Internet
2 Data Link	Frames and MAC addresses	Network Access
1 Physical	Signal transmission	Network Access

Expected observation

Event	Interpretation
ARP Request/Reply	Resolves the target IPv4 address to a MAC address
ICMP Echo Request	Tests reachability from source to destination
ICMP Echo Reply	Confirms the target can respond
Ethernet/IP headers	Show link-layer and network-layer encapsulation

Interpretation / Analysis

- The switch forwards Ethernet frames using MAC addresses while IP identifies communicating endpoints.
- An initial ping can cause ARP resolution before ICMP messages are exchanged.
- OSI is a conceptual seven-layer model; TCP/IP is the practical protocol-suite model.

Evaluation Against a No-Code Solution

- Ease of creation: determine how much is configured visually rather than programmed.
- Data and integration: identify where records, connections or services are created and managed.
- Governance: consider role permissions, traceability and safe data handling.
- Limitations: identify any feature that may require paid access, external services or custom logic.

Verification / Test Cases

Suggested Test Record

Test	Input / action	Expected observation
Normal test	Valid sample values and supported action	Expected successful output
Boundary test	Missing value, invalid input or empty data	Validation or clear empty state
Access test	Unapproved user attempts restricted action	Access limited according to role
Repeatability	Re-run same valid example	Consistent output or explain differences

Observation Notes

During the practical session, replace the illustrative outcomes with actual observed results. Include the software/platform name, version if known, screenshots, and any errors encountered.

Results

The topology and layer mapping illustrate how data is encapsulated, transmitted, received, and interpreted. Ping and simulation traces are the suggested evidence; actual results must be recorded after running the simulator.

Conclusion

Layered models simplify network analysis and troubleshooting. The simulator demonstrates how ARP, Ethernet, IPv4 and ICMP cooperate to provide host-to-host connectivity.

No-Code Learning Outcome

This experiment links the technical concept to a practical no-code application development workflow. It highlights visual configuration, user experience, data handling and safe validation rather than treating the exercise as conventional full-code software development.

Precautions

- Use only fictional sample records for classroom demonstrations; avoid exposing passwords, tokens, employee information or real contact details.
- Record your own screenshots, observed outputs and deviations after executing the experiment. The examples below are models, not verified live results.
- Check access rights before sharing or publishing an app, report, workflow or externally reachable service.